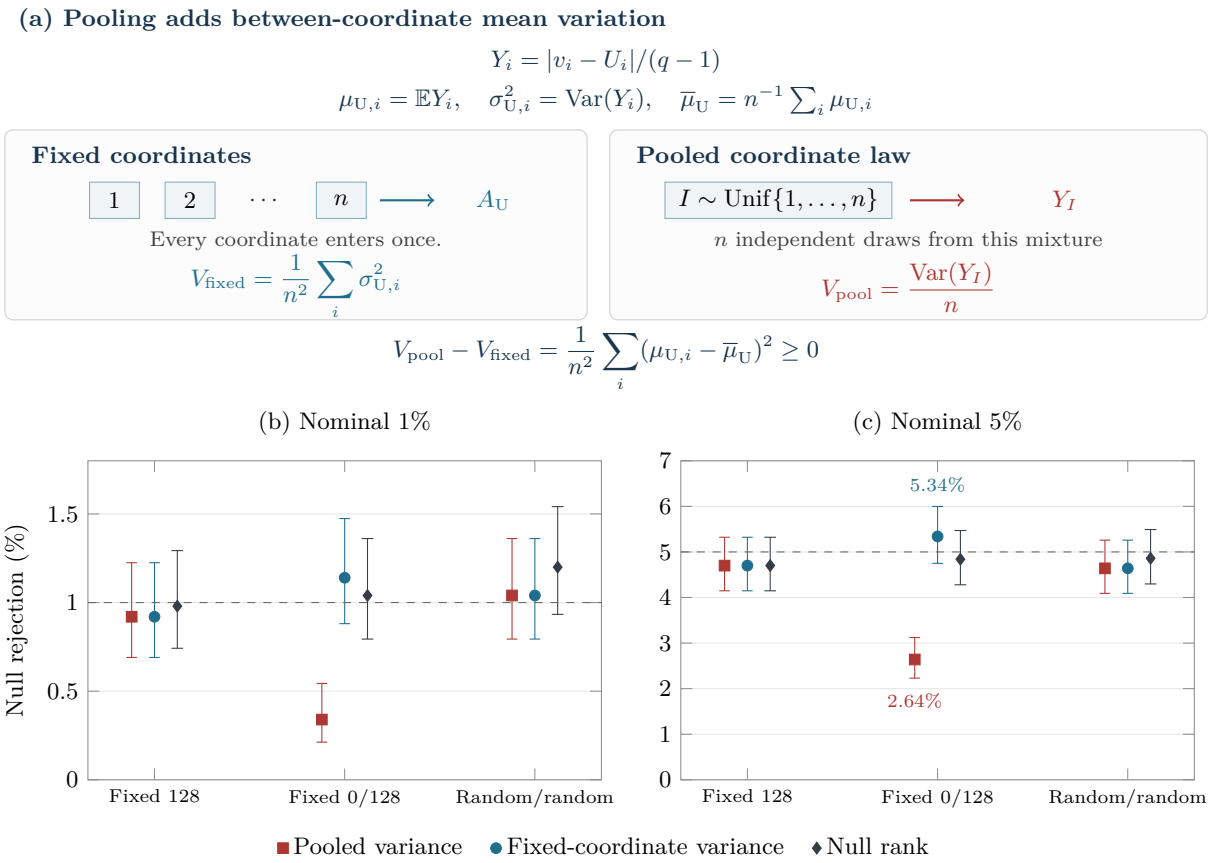
## UACI pooling and fixed-coordinate calibration




Unified average changing intensity (UACI) under fixed coordinates and the effect of pooling conditional laws. Panel (a) gives the exact variance decomposition for a fixed image $\boldsymbol{v} = (v_1, \dots, v_n)$ with $v_i \in \{0, \dots, q-1\}$ and mutually independent uniform pixels $U_i$ on the same alphabet. Here $A_{\mathrm{U}} = n^{-1}\sum_i Y_i$, $\mathrm{Unif}\{1, \dots, n\}$ denotes the uniform distribution on the coordinate labels, and $I$ is independent of the pixels. The pooled experiment averages $n$ independent draws from the mixture law of $Y_I$. Panels (b,c) show 5000 null evaluations for each of three $64 \times 64$ designs at $q = 256$. The first two legend entries denote Gaussian rules using the pooled variance [1] and the fixed-coordinate variance. Fixed 0/128 denotes the checkerboard. The pooled and fixed-coordinate variances agree for the constant image. The random/random control uses two mutually independent uniform images, and both Gaussian rules use their common variance. Dashed lines mark the nominal levels. The panels have different vertical scales. Error bars are Wilson 95% intervals, conditional on the realized independent bank of $L = 9999$ reference statistics for the independent-null rank rule.

# Reference


[1] Y. Zhang, Statistical test criteria for sensitivity indexes of image cryptosystems, *Information Sciences* 550 (2021) 313–328. DOI 10.1016/j.ins.2020.10.026.